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Optimal Placement of Laying Hen House Temperature Sensors Based on Genetic Algorithm

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ABSTRACT The arrangement of temperature sensors in most existing large-scale laying hen houses is random or placed according to the experience of breeders. However, this can't achieve accurate monitoring of the henhouse. The temperature in the laying hen houses cannot be uniformly controlled, leads to the reduction of production efficiency of laying hens. In this paper, aiming at the placement of temperature sensors in laying hens' houses, a placement optimization method was proposed. Firstly, the correlation coefficient of sensors is calculated to eliminate redundant sensors. Then, all the remaining sensors are arranged and combined. Finally, taking the grey correlation degree of each combination as the objective function, the dual-structure coding genetic algorithm is used to optimize the sensor combination. The strategy was tested in a large hen house. When the initial deployment of 81 sensors is reduced to at least 3, the established target can still be achieved, and the position of the target sensor can be calculated. This strategy not only meets the goal of accurately monitoring the hen house temperature, but also saves the hardware cost, which has important application value.

INDEX TERMS Dual-structure coding, genetic algorithm, grey correlation degree, large laying hen house, optimal sensor placement.

I. INTRODUCTION

Modern laying hen houses mostly use high-density stacked cage raising technology, and the key of stacked cage raising lies in the control of indoor environment [1]. Temperature has a great influence on the performance of laying hens. Too high or too low temperature in poultry house is not conducive to their growth. Low temperature will cause hens to consume too much energy to maintain their needs, thus reducing the number of eggs laid, the frequency of laying eggs and their immunity. If the temperature is too high, it is easy to cause diseases and reduce the laying rate and food intake of hens [2]. Therefore, hen farms need to take corresponding measures to control the temperature within an appropriate range. However, due to the large span and high feeding density, it is difficult to control the internal environment temperature uniformly and properly. Moreover, the data measured at a specific location cannot completely represent the temperature of the environment in the whole laying hen house. If a large number of sensors are placed, the temperature change in the laying hen house can be monitored more comprehensively, but this will not only increase the cost, but also increase the complexity of the system. If the position of temperature sensor is optimized, the cost and workload of measuring and processing a large number of redundant sensor data can be greatly reduced. Proper sensor position will give correct feedback, so that the laying hen house will maintain good comfort and provide a comfortable living environment for hens. Therefore, the purpose of this paper is to optimize the monitoring of temperature by using the least number of sensors and arranging them at the best position.

Optimal placement of sensors is a combinatorial optimization problem, which means that a small number of sensors are placed at appropriate positions to obtain enough environmental information. At present, there are not many applications of sensor layout optimization in laying hen house.

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Huang and Li [3], [4], aiming at the sensor combinatorial optimization problem of structural damage diagnosis, took the largest non-diagonal element of the modal confidence matrix as the objective function, and solved the constrained combinatorial optimization problem by using dual-structural coding genetic algorithm. Furthermore, the information obtained from the predicted data by information entropy is taken as the main criterion for evaluating each sensor, so as to select the best position of the sensor [5], [6]. Liu *et al.* [7] put forward the fuzzy matter-element analysis method based on multi-objective decision, and divided the monitoring points into seven categories according to the comprehensive correlation degree of the monitoring data of different monitoring points, thus realizing the optimization of the sensor layout in the refrigerated compartment. Lu and Zhang [8] used the improved particle swarm optimization algorithm, applied the optimal arrangement of humidity sensors in ancient buildings, took the maximization of temperature difference as the objective function, and determined the optimal number of sensors according to the increasing gradient of fitness value. According to the data collected by arranging a large number of sensors, the sensors are grouped by clustering algorithm, and the information loss is calculated by Pareto principle to determine the optimal sensor [9]–[12]. Lin *et al.* [13] applied the deep reinforcement learning to the optimal position of the plantar pressure sensor, and found the optimal position of the plantar pressure sensor without any prior knowledge of the anatomical region of the foot. Uyeh *et al.* [14] used reinforcement learning algorithm to determine the best sensor position of temperature and humidity sensors in greenhouse, which reduced the number of sensors from 56 to 10. Kang *et al.* [15] applied neural network modeling to optimize the sensor position for measuring flue gas in industrial pipeline.

Most of the above are optimized based on the data collected by randomly or uniformly arranged sensors. Some scholars construct temperature field or humidity field by arranging a small number of sensors, and further optimize the temperature and humidity sensors according to the interpolation or simulation results. Curi measured the temperature and humidity of broiler house and combined them into the temperature and humidity index. By using Kriging interpolation to the temperature and humidity index in broiler house, three best positions were selected in each different type of broiler house [16]. Cuadrado *et al.* combined the inverse heat conduction method with genetic algorithm optimization based on kriging interpolation to determine the position of the sensor in microchip [17]. Fu *et al.* [18] proposed a greedy thermal sensor placement algorithm to arrange the thermal sensors in the chip. The position optimization of the sensor can also be realized by using hydrodynamic simulation [19]–[21]. Loong *et al.* applied the fuzzy logic simulation to obtain the best location for sensor installation with the most faults detected [22]. Arnesano *et al.* [23] and Lee *et al.* [24] used error-based and entropy-based methods to select the sensor closest to the indoor average temperature and the sensor most affected by the external environment. However, they only sort the sensor combinations according to the average, standard deviation, outliers and z-index, and they didn't choose the optimal number of sensor combinations. Therefore, a better measurement index is needed to quantify the sensor combination, so as to select the optimal sensor combination.

The above sensor optimization methods have been well proved in various structures and special environments. However, due to the special environment of the laying hen house, some methods are not suitable for the layout optimization of sensors in the laying hen house, and some methods have not been applied to the layout optimization of temperature sensors. There is little research on the optimal layout of sensors in the laying hen house, most of them are arranged in same horizontal plane, without considering the spatial distribution of temperature. Because the environment in the laying hen house is uncertain, it is necessary to consider not only the spatial distribution of temperature, but also the change of temperature in time. Genetic algorithm with dual-structure coding has only been applied in the optimal configuration of sensors in the field of structural health monitoring, but has not been verified in the optimization of temperature sensors. Moreover, the genetic algorithm with dual-structure coding can't solve the optimization problem of temperature sensors in laying hen house well. We need an index to measure the quality of each sensor combination as the objective function of the genetic algorithm. Therefore, it is necessary to combine the grey correlation degree with the genetic algorithm of dual-structure coding, and use the grey correlation degree of sensor combination as the measurement index to optimize the placement of temperature sensors in laying hen house.

In order to optimize the position and number of temperature sensors in the henhouse, this paper combines the genetic algorithm of dual-structure coding with the method of grey correlation degree, and provides the best number and position of temperature sensors for the input part of the temperature control system in the henhouse. In the process of optimization, the following problems need to be solved: (1) Establishment of the structure of multi-layer cage laying hen house and the initial layout model of temperature sensors. (2) Optimization of the number and location of temperature sensors. (3) Selection and verification of optimization methods.

According to the above problems, this paper puts forward a model of hen house structure and temperature sensor arrangement, aiming at the least number of temperature sensors in multilayer cage hen house, taking a large modern multilayer cage hen house as the research object, eliminating some redundant sensors by correlation coefficient method. Considering the two factors of sensor position and number, and combining genetic algorithm with grey correlation degree, solving the problem of the number and position of temperature sensors in hen house, and improving the efficiency and economy of temperature control system in hen house.

II. HENHOUSE STRUCTURE AND SENSOR DEPLOYMENT MODEL

In section II, firstly, the structure and characteristics of the multi-layer cage laying hen house are introduced, and then the temperature sensor deployment model is constructed according to the characteristics of the hen house, and the original air temperature data are collected according to the model.

A. LAYER HOUSE STRUCTURE AND ITS CHARACTERISTICS

In this paper, a laying hen farm was selected as the research base, located in Shijiazhuang City, Hebei Province, at 37 75′ north latitude and 115 29′ east longitude. March to April in Shijiazhuang is spring, dry and windy; June to August is summer, hot and rainy; September to November is autumn, mild and cool; December to February is cold with little sunshine and little rain and snow in winter. The annual average temperature is 12.5°C, the coldest in January and the monthly average temperature is −3.9°C. It is hottest in July, with a monthly average temperature of 26.5°C. The monitored single laying hen house has a length of 78m, an inner width of 15.5m, a double slope roof, a low eaves height of 3.9m, a ridge height of 5.4m, and four floors and five rows of hen cages, as shown in Fig. 1. The breeding quantity is about 37,000. The laying hen house is equipped with ventilation and cooling equipment such as automatic feeding equipment, automatic drinking water system, automatic dung cleaning system, egg collecting system, fan and wet curtain, lighting system and automatic control system. It is a fully enclosed laying hen house. The farm has initially achieved the automation level of large-scale laying hen farms, but there are also some common problems in the digital management of facility farms, such as the backward management of the internal environment control facilities of laying hen houses.

In summer, a large amount of heat is accumulated in the layer house. In order to increase the ventilation rate and make better use of the evaporation and air cooling effect of wet curtain, it is necessary to increase the wind speed and promote the air flow in the house. Wet curtain is installed at the air inlet of the layer house to cool down, and 2∗8 exhaust fans are installed at the far end of the layer house. The exhaust fans exhaust outwards to generate negative pressure in the layer house. Hot air enters through the air inlet of wet curtain, and after evaporation and cooling by wet curtain, the temperature drops and flows along the longitudinal direction of the layer house. Setting temperatures for starting exhaust fans and water curtains in layer house in summer are as follows: the fans turn on or off according to the change of indoor temperature. When the temperature continues to rise, the number of fans will increase or the ventilation time will increase, and when the temperature continues to drop, the fans will be turned off; The wet curtain is manually opened at 10: 00 a.m. and closed at around 4: 00 p.m. every day.

B. SENSOR DEPLOYMENT MODEL

In summer (August 2020), temperature sensors were used to measure the temperature of 81 fixed points in the laying hen house every hour, as shown in Fig. 2. It can be seen from the Fig. 3 that although the temperature changes of different sensors are different, the overall change trend is consistent. Within a whole day, there will be a peak in the temperature curve of the henhouse, and the highest temperature of the henhouse is from 10: 00 to 16: 00 every day. Since the fan and wet curtain are turned off from 1: 00 to 10: 00, as the sun rises and the laying hens are active, a lot of heat is generated. With the opening of the wet curtain, the indoor temperature gradually decreases. Although there is a control system in the henhouse to adjust the temperature, the temperature at different positions in the henhouse is still different at the same time, and the temperature difference can reach about 5°C, and the temperature difference at different heights can also reach about 3°C, which proves that the air temperature distribution in the henhouse is uneven in horizontal and vertical directions. Moreover, the temperature of the same sensor fluctuates greatly in one day, and the temperature difference can reach about 7°C. Therefore, it is necessary to arrange the sensor position reasonably.

Because of the uneven distribution of air temperature in the henhouse, in order to obtain air temperature data in different areas in the henhouse, it is necessary to distribute air temperature sensors in different positions of the henhouse. If the number of temperature sensors arranged is too small, and the locations are not appropriate, the monitored temperature data is not enough to represent the temperature of the whole environment of the henhouse. If there are too many sensors, not only the local base station can't bear it, but also a lot of data may be lost and the cost will increase. Considering the possibility of installation position and the uniformity of layout, 81 points are reasonably arranged in the laying hen house. These points are arranged outside the hen cages, close

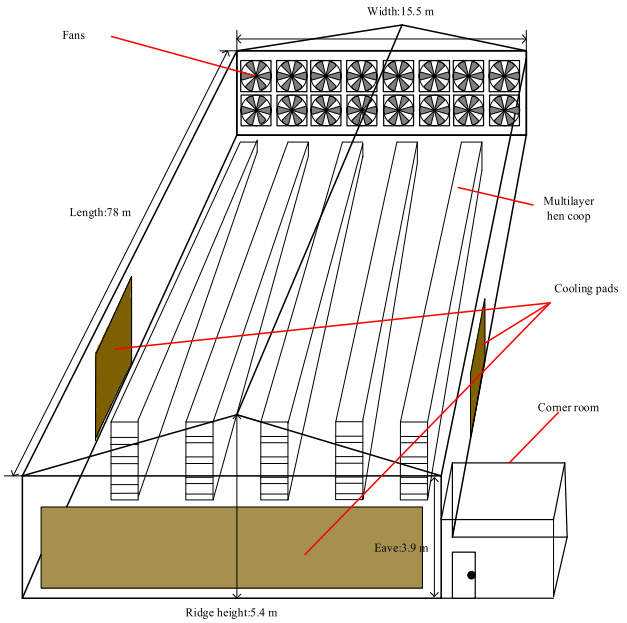


FIGURE 1. Schematic diagram of experimental laying hen house.

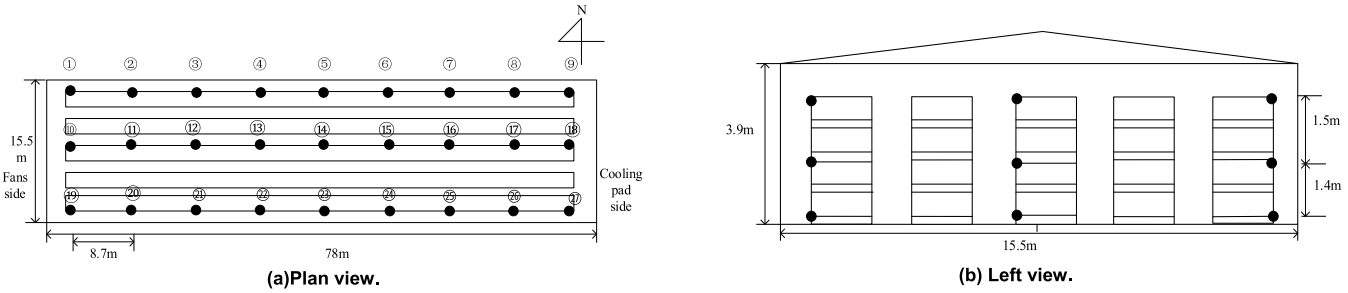


FIGURE 2. Arrangement position of temperature sensor in henhouse: (a)Plan view; (b) Left view.

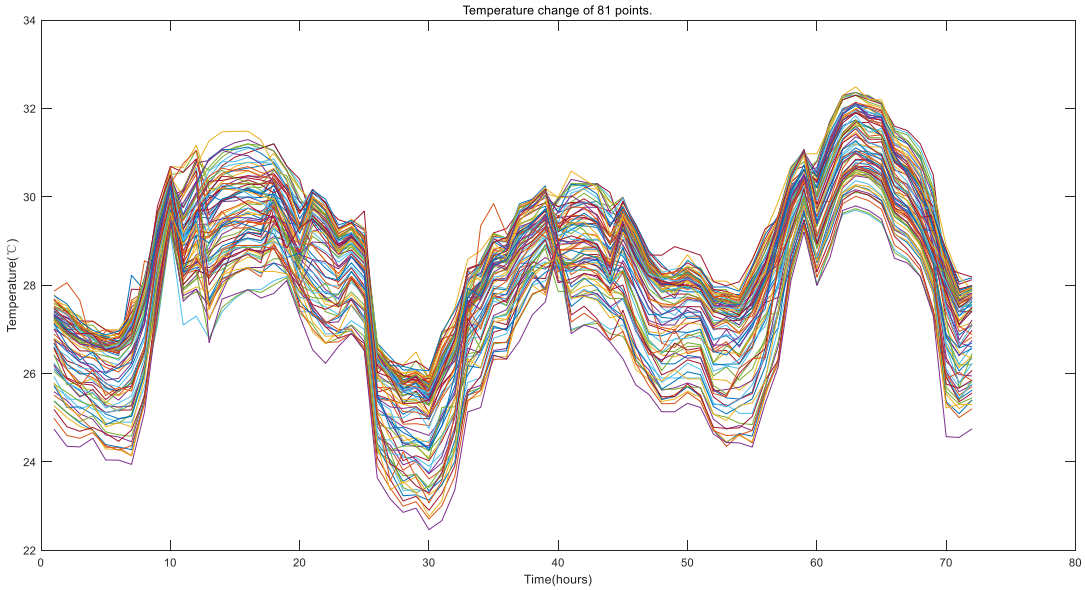


FIGURE 3. The temperature change curve of 72 hours in the laying hen house.

to the first aisle, the third aisle and the sixth aisle, and a position is arranged every 8.7 meters (Figure 2a). At the same time, the laying hen house is divided into three parts in the horizontal direction, that is, planes with measuring heights of 0.7m, 1.4m and 2.9m (Figure 2b). Wireless temperature sensors are arranged for synchronous measurement at points and the data of other points are obtained by interpolation. Temperature sensors are set to collect data once every hour to realize comprehensive monitoring of the temperature in the laying hen house. The temperature monitoring equipment is a temperature recorder developed by Ying Ba Company, which supports data recording and cloud storage. Other related equipment parameters are shown in Table 1.

TABLE 1. Parameters of temperature sensor equipment.

size	113mm*64mm*15mm
weight	120g
measuring range	-40°C~85°C
operation temperature	-20°C~50°C
measuring accuracy	±0.5°C
battery capacity	1750mAh
communication technology	General Packet Radio Service

III. OPTIMIZATION STRATEGY BASED ON GENETIC ALGORITHM

The air temperature inside the laying hen house was obtained at 81 points to determine the best sensor point for accurately monitoring the internal environment. Based on the long-term data measured and data interpolation in the laying hen house, the combination of dual-structure coding genetic algorithm and grey correlational degree is used to evaluate the combination of all sensor positions to determine which combination can best represent the overall environment of the laying hen house. The experimental process is shown in flow chart of Fig. 4.

A. GENETIC ALGORITHM WITH DUAL-STRUCTURE CODING

The placement optimization of sensors in the laying hen house can be transformed into combinatorial optimization.

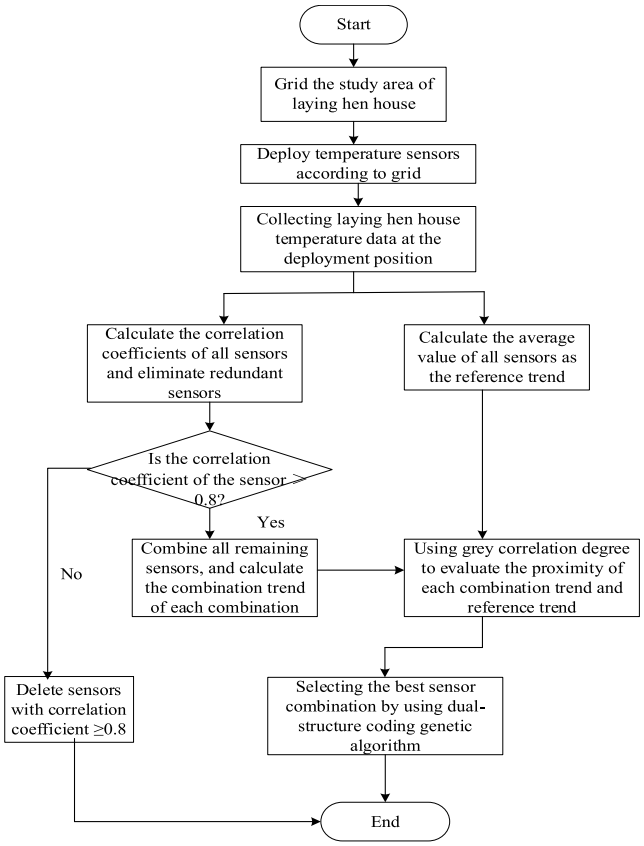


FIGURE 4. Optimization flow chart of temperature sensor in laying hen house.

There are many methods to solve combinatorial optimization, including traditional algorithm, greedy algorithm, random algorithm and so on. Artificial intelligence algorithms include neural network, simulated annealing algorithm and genetic algorithm. When the number of sensors increases, the combination will increase explosively, so some artificial intelligence algorithms are needed to solve this problem. Compared with other algorithms, genetic algorithm has the advantages of fast convergence, fast searching speed, and not easy to fall into the optimal solution. Therefore, this paper chooses genetic algorithm to solve the optimal sensor position combination. However, the traditional genetic algorithm uses binary coding, and the number of sensors will change because of changing the number of gene coding 1. Traditional genetic algorithms usually use penalty strategy to meet the constraint condition of constant number of sensors, but improper setting of penalty function will cover up the optimization of objective function. Therefore, this paper adopts the dual-structure coding genetic algorithm, which can ensure that the number of sensors remains unchanged during the crossover and mutation operation.

We can regard the trend of average temperature change in the house as a target curve that changes with time, and the trend of average temperature change of different sensor combinations as a set of curves. With the gray correlation value of the temperature change curve of each sensor combination and the target curve as the constraint condition, we can calculate the sensor combination with the largest gray correlation degree in the population through each iteration of the genetic algorithm with dual-structure coding, that is, the sensor combination closest to the average temperature change in the house. That is to say, a group of feasible solutions of sensor combination optimization problem is regarded as an individual, and a population is composed of a certain number of feasible solutions. By calculating the grey correlation degree between each individual and the reference trend as the fitness function, and taking the grey correlation degree as the fitness degree of each individual in the population, the advantages and disadvantages of biological individuals are evaluated according to the fitness degree. Genetic operations with biological significance, such as natural selection, crossover and mutation, are used to generate an evolved new generation of population, so that the population is updated and iterated until the optimal individual is produced [25]. The flow of genetic algorithm with dual-structure coding is as follows.

TABLE 2. Coding of dual structure method.

Additional code	x(1)	x(2)	…	x(i)	…	x(n)
Variable code	$y_a(1)$	$y_a(2)$	…	$y_a(i)$	…	$y_a(n)$

1) Population initialization. The mathematical model of optimal placement of sensors is a 0-1 programming problem. If the traditional binary coding method is adopted, the number of sensors will be changed due to changing the number of gene codes 1 during the crossover mutation operation. In this paper, the dual-structure coding is adopted to satisfy the constraint condition that the number of sensors is unchanged, that is to say, the number of sensors in each population is consistent [26]. The dual-structure coding method is shown in Table 2. Individual chromosomes are composed of additional code and variable code: the additional code indicates each position where the sensor can be arranged, and the variable code is composed of 0/1 combination, indicating whether the sensor is arranged at this position (value is 1, that is, arrangement; Is 0, it is not arranged). When encoding an individual, the additional code is generated by shuffling, and the corresponding variable code is generated randomly, which constitutes an individual's dual-structure encoding.
2) Fitness calculation. According to the fitness value, the optimal combination under each sensor number is calculated.
3) Select operation. According to the fitness of individuals in the population, individuals with high fitness are selected from the current population by roulette. The higher the fitness, the greater the chance of

being selected. Because of the randomness of roulette wheel selection, it is impossible to ensure that the best individual enters the next round of competition. In order not to lose the best solution, the elite strategy is implemented in the selection process, that is, the best individual is retained. After the algorithm completes crossover and mutation operations, the worst individual in the next generation population is replaced by the best individual to ensure that the population does not deteriorate.

4) Crossover operation. In the dual structure coding, if the common crossover operator is used, the additional codes that produce new individuals will be repeated, thus producing contradictory individuals. PMX operation only aims at the additional codes of individuals [27], and the specific steps are as follows:
Step1: Select the starting and ending positions of several genes in a pair of chromosomes randomly. The segment in the middle of the starting and ending position of chromosome is called a matching portion. The length of two chromosome matching portions must be the same.

$$parent1 = \begin{bmatrix} additional\ \ code: & 1 & 2 & 3 & |4 & 5 & 6| & 7 & 8 & 9 \\ variable\ \ code: & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$parent2 = \begin{bmatrix} additional\ \ code: & 1 & |3 & 8 & 9| & 2 & 6 & 5 & 7 & 4 \\ variable\ \ code: & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

Step2: Interchange matching portions in two chromosomes.

$$child1 = \begin{bmatrix} additional\ \ code: & 1 & 2 & 3 & |3 & 8 & 9| & 7 & 8 & 9 \\ variable\ \ code: & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$child2 = \begin{bmatrix} additional\ \ code: & 1 & |4 & 5 & 6| & 2 & 6 & 5 & 7 & 4 \\ variable\ \ code: & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

Step3: Keep the matching portions unchanged, look for duplicate values in the segments that have not been swapped, and then find the corresponding elements to replace them in the replaced parts of the parent. Finally, it is ensured that a new pair of offspring genes will not conflict.

$$child1 = \begin{bmatrix} additional\ \ code: & 1 & 2 & 3 & |5 & 6 & 4| & 7 & 8 & 9 \\ variable\ \ code: & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$child2 = \begin{bmatrix} additional\ \ code: & 1 & |9 & 5 & 8| & 2 & 6 & 3 & 7 & 4 \\ variable\ \ code: & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

5) Mutation operation. The reverse mutation operator is adopted, that is, two mutation points are randomly generated for the parent, and only the ascending additional codes between the two points are rearranged in

$$\text{NewChrom} = \begin{bmatrix} 3 & 5 & 7 & |8 & 6 & 2 & 9| & 1 & 4 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} Child$$

reverse order, while the descending variable codes are unchanged.

$$\text{OldChrom} = \begin{bmatrix} 3 & 5 & 7 & |9 & 2 & 6 & 8| & 1 & 4 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} Parent$$

6) Termination of judgment. If the number of iterations is satisfied or the optimal solution has not changed for several generations, the algorithm is terminated, otherwise, the second step is returned.

B. OPTIMIZATION STRATEGY

The basic idea of grey correlational analysis is to judge whether the connections between different sequences are closely related according to the similarity of geometric shapes of sequence curves. The closer the curve shape is, the greater the correlation degree of the corresponding index series is [28], [29]. The greater the difference of curve shape, the smaller the correlation degree of the corresponding index series. Assuming that the average value of data measured by all sensors represents the whole environment, it is used to select the best combination of sensor positions representing the whole environment in the henhouse. At first, the Average Temperature of Overall is calculated by averaging the air temperature data measured by all sensors. Average temperature of combination is calculated by averaging the air temperature data measured by all sensors after eliminating redundancy. The degree of closeness between each combination trend and the reference trend is evaluated by the grey correlation degree. The larger the grey correlation degree corresponding to each combination trend, the closer the temperature change of this combination is to the temperature change of the reference trend, that is, it can represent the average temperature change of the whole environment in the henhouse.

1) Determine the reference Trend T_r as a reference series, and the combined trend T_c as a comparison series, and form a matrix:

$$\begin{aligned} &\left(T_r, T_{c,1}, \cdots, T_{c,m}\right) \\ &= \begin{bmatrix} T_r(1) & T_{c,1}(1) & \cdots & T_{c,j}(1) & \cdots & T_{c,m}(1) \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ T_r(i) & T_{c,1}(i) & \cdots & T_{c,j}(i) & \cdots & T_{c,m}(i) \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ T_r(n) & T_{c,1}(n) & \cdots & T_{c,j}(n) & \cdots & T_{c,m}(n) \end{bmatrix} \end{aligned} \tag{1}$$

In equation (1), t ($t = 1, 2, \cdots n$) is the dimension of matrix, which represents the sensor acquisition time in this paper; j ($j = 1, 2, \cdots, m$) is the number of combined trends; $T_r(i)$ represents the data of reference trend T_r at time i; $T_{c,j}(i)$ represents the data of the j group combination trend at time i.

2) In order to eliminate the dimension and unify the quantity, the data series is dimensionless by using the "extreme difference" processing method. Poor treatment:

$$T_{c,j}(i) = \frac{T_r - \min(T_r)}{\max(T_r) - \min(T_r)} \tag{2}$$

3) Calculate the absolute difference between the system characteristic (standard) sequence and the correlation factor behavior (comparison) sequence one by one. That is, $\Delta_{r,c} = \left|T_r(i) - T_{c,j}(i)\right|$ $(i = 1, 2, \cdots, n; j = 1, 2, \cdots, m)$
4) Calculate $\min\limits_{j}\min\limits_{i}\Delta_{r,c}$ and $\max\limits_{j}\max\limits_{i}\Delta_{r,c}$. Among of them, $\min\limits_{j}\min\limits_{i}\Delta_{r,c}$ is the minimum difference between T_r and $T_{c,j}$ at time k. $\max\limits_{j}\max\limits_{i}\Delta_{r,c}$ is the maximum difference between T_r and $T_{c,j}$ at time k.
5) Calculate the correlation coefficient between $T_{c,j}$ and T_r.

$$\zeta_j(i) = \frac{\min\limits_{j}\min\limits_{i}\left|T_r(i)-T_{c,j}(i)\right| + \rho \max\limits_{j}\max\limits_{i}\left|T_r(i)-T_{c,j}(i)\right|}{\left|T_r(i)-T_{c,j}(i)\right| + \rho \max\limits_{j}\max\limits_{i}\left|T_r(i)-T_{c,j}(i)\right|} \tag{3}$$

In equation (3), ρ is the resolution coefficient, $\rho \in [0, 1]$, If it is smaller, the difference between correlation coefficients will be larger, and the discrimination ability will be stronger. Here, ρ is taken as 0.5. [30]
6) The correlation coefficient between $T_{c,j}$ and T_r can be expressed as follows:

$$\gamma(T_r, T_{c,j}) = \frac{1}{n}\sum_{i=1}^{n}\gamma(T_r(i), T_{c,j}(i)) \tag{4}$$

In equation (4), $\zeta_j(i) = \gamma(T_r(i), T_{c,j}(i))$.

By calculating the grey correlation degree of each combination under each sensor number, the optimal sensor combination under each sensor can be obtained, but the obtained grey correlation degree value is calculated relative to other combinations under the same number, so the grey correlation degree of combinations under different numbers cannot be compared. Therefore, we should combine the optimal sensors with different numbers, and calculate the grey correlation degree between the optimal combination trend and the reference trend under each number, so as to determine the optimal number of sensors to be placed.

IV. RESULTS

A. ANALYSIS OF ENVIRONMENTAL TEMPERATURE DATA

Fig. 5 shows the temperatures of the top 27 sensors at 10:00, 14:00, 18:00 and 22:00 on August 8th. It can be concluded that when the fan and wet curtain are not turned on, the temperature in the middle of the laying hen house is higher than that at both ends, which may be due to less air flow in the middle than at both ends and a large amount of heat accumulated. With the opening of the fan and wet curtain, the temperature in the laying hen house increases gradually from the wet curtain end to the fan end, and the maximum increase range is about 3°C. As the sun rises, the temperature in the south wall is slightly higher than that in the north wall. As the sun goes down, the temperature in the north and south columns tends to be consistent. Fig. 6 shows the temperature distribution of three sensors with the same horizontal and vertical coordinates and different heights within 24 hours on August 8. The temperature distribution of sensors 11, 38 and 65 in the height direction basically accords with the distribution of middle height and slightly lower up and down. In most time periods, the middle layer temperature is higher than the bottom layer temperature and higher than the top layer temperature.

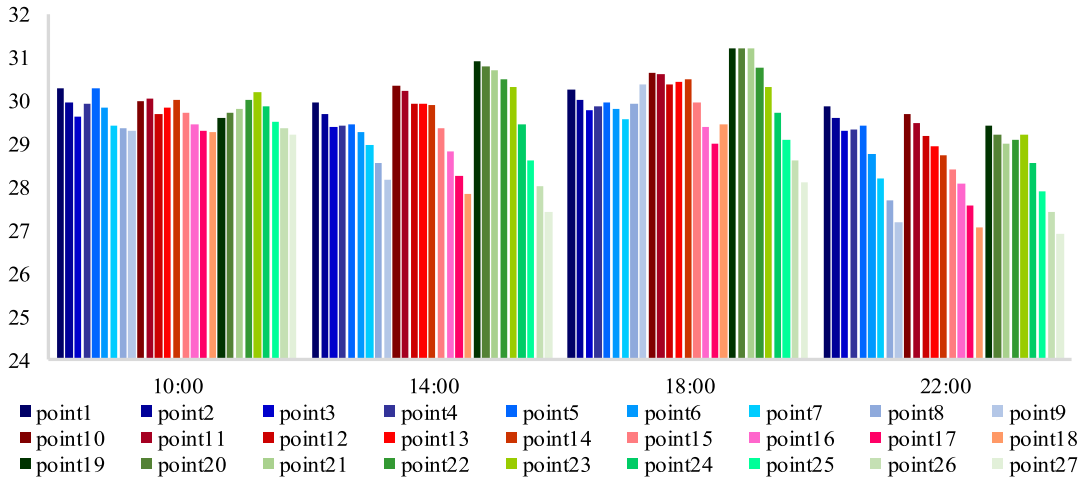


FIGURE 5. Temperature distribution map of henhouse in different positions on the top floor.

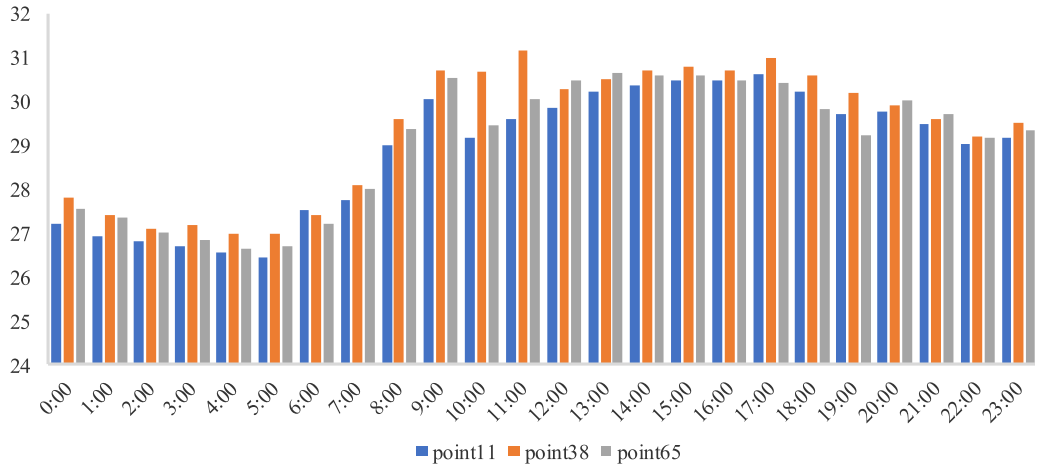


FIGURE 6. Temperature comparison vertically in laying hen house.

B. ELIMINATION OF REDUNDANT SENSORS BY CORRELATION COEFFICIENT

The temperature in the whole henhouse is closely related to the cooling system, therefore, the temperature in the same laying hen house has the same change trend. In other words, the temperature change trend of each sensor is basically related. However, because the installation position of the fan and wet curtain is fixed, and the laying hen house has a large span, the temperature distribution is uneven. If the non-correlation between a sensor and other sensors is high, it means that the data of this sensor can' t be estimated by the data of any neighboring sensor, so it is necessary to place a sensor at this position. Furthermore, if the correlation between the temperature measured by each sensor and others is small, it shows that a set of data measured by all sensors can well characterize this distribution state. On the contrary, if the data measured by two or more sensors have high correlation, the arrangement of sensors is redundant. The correlation intensity adopts the correlation standard of linear correlation analysis, that is, there is a significant relationship between the correlation degree and the correlation coefficient. Generally speaking, the magnitude and degree of correlation of correlation coefficient have the corresponding relationship of the following formula (5).

$$\begin{cases} 0 < \mathrm{r} \leq 0.3 & \text{slightly correlation,} \\ 0.3 < r \leq 0.5 & \text{moderately correlation} \\ 0.5 < r \leq 0.8 & \text{significant correlation} \\ 0.8 < r \leq 1 & \text{fiercely correlation} \end{cases} \tag{5}$$

TABLE 3. Ranking of correlation coefficients of 81 sensors.

Ranking	Value	ID	Ranking	Value	ID	Ranking	Value	ID	Ranking	Value	ID
1	0.6662	44	21	0.8277	66	41	0.8490	73	61	0.8971	52
2	0.6755	5	22	0.8278	39	42	0.8501	74	62	0.8982	35
3	0.7139	19	23	0.8282	55	43	0.8606	7	63	0.9034	13
4	0.7153	32	24	0.8295	11	44	0.8694	27	64	0.9223	15
5	0.7192	20	25	0.8319	29	45	0.8694	24	65	0.9258	36
6	0.7197	21	26	0.8332	77	46	0.8720	51	66	0.9329	38
7	0.7366	46	27	0.8334	56	47	0.8726	17	67	0.9339	67
8	0.7418	47	28	0.8340	22	48	0.8727	61	68	0.9378	40
9	0.7596	59	29	0.8348	33	49	0.8735	34	69	0.9394	42
10	0.7611	31	30	0.8374	75	50	0.8737	9	70	0.9450	45
11	0.7665	4	31	0.8381	23	51	0.8739	8	71	0.9457	14
12	0.7910	3	32	0.8415	76	52	0.8741	16	72	0.9535	69
13	0.7988	30	33	0.8416	64	53	0.8763	78	73	0.9581	54
14	0.8036	1	34	0.8431	37	54	0.8800	18	74	0.9590	41
15	0.8040	2	35	0.8435	48	55	0.8840	26	75	0.9626	62
16	0.8086	57	36	0.8452	50	56	0.8864	43	76	0.9725	63
17	0.8163	58	37	0.8459	65	57	0.8872	70	77	0.9746	68
18	0.8168	28	38	0.8464	60	58	0.8877	25	78	0.9799	80
19	0.8175	6	39	0.8466	10	59	0.8921	53	79	0.9801	71
20	0.8242	12	40	0.8469	49	60	0.8940	79	80	0.9923	72

TABLE 4. Optimal sensor combination under different numbers.

Quantity of sensor combinations	Best combination of sensors												
1	4												
2	32	47											
3	21	47	59										
4	21	30	47	59									
5	20	21	32	47	59								
6	21	31	32	46	47	59							
7	19	21	30	44	46	47	59						
8	4	19	20	30	44	46	47	59					
9	4	20	21	30	32	44	46	47	59				
10	4	19	20	21	30	32	44	46	47	59			
11	4	5	19	20	21	30	31	44	46	47	59		
12	3	4	5	19	20	21	30	31	44	46	47	59	
13	3	4	5	19	20	21	30	31	32	44	46	47	59

Calculate the sum of correlation coefficients between each sensor and other remaining sensors, and then arrange all sensors from small to large according to the sum and average of correlation coefficients, so as to eliminate some sensors with large correlation coefficients. In this way, some redundant sensors are deleted, which is beneficial to improve the efficiency of subsequent optimization. Firstly, calculate the correlation coefficient among all sensors, and select the value with the smallest correlation coefficient, which is calculated as No.44 and No.5. Then, select the sensor with the smallest average correlation coefficient, calculate it as the No.44 sensors. Delete the coefficient value between it and other sensors in the correlation matrix, and then continue to select the next best sensor after the remaining correlation matrix is reduced in dimension once, thus gradually calculating to the last sensor. By sorting and recording the obtained sensors, 81 position sensors are finally sorted as shown in Table 3. Select sensors smaller than 0.8 for further optimization (the first 13 sensors in Table 3).

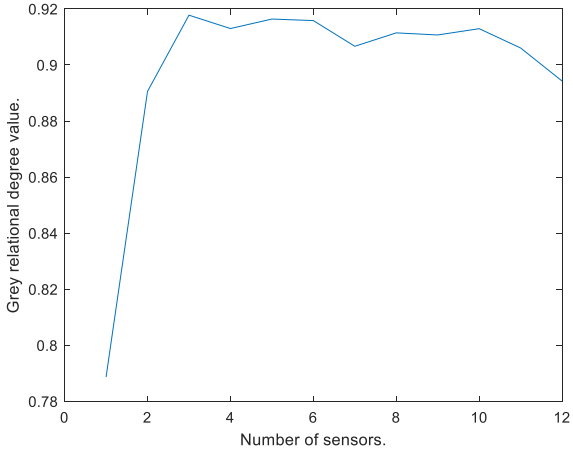


FIGURE 7. Grey correlation degree between the best sensor combination trend and reference trend under different number.

Since each gene in an individual represents a different sensor, the number of genes in an individual is set to 13. The number of sensors in the combination is controlled by controlling the number of "1" in the gene. If the population is too small, it is obvious that inbreeding will occur, resulting in morbid genes. So that the population evolution can't produce the expected quantity according to the pattern theorem. Because the population size is too large, the result is difficult to converge, resources are wasted, and the stability is reduced, so the population size is set to 50. The evolution algebra is too small, and the population evolution is immature, so it is not easy to converge. The evolution algebra is too large, the population evolution is too premature to converge, and it is meaningless to continue evolution, which will only increase time expenditure and waste of resources. The maximum number of iterations is set to 200 times, and the termination condition is that the maximum evolutionary algebra or the optimal solution remains unchanged for several generations. The crossover probability is too small to effectively update the population, resulting in unconvincing convergence results. The crossover probability is too large, similar to the mutation probability, which easily destroys the existing optimal solution, increases the randomness and

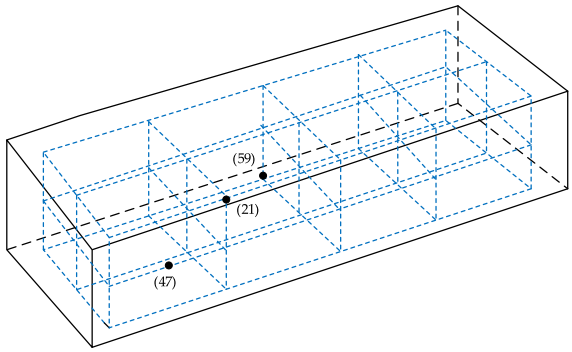


FIGURE 8. The position of the best sensor combination in the henhouse.

TABLE 5. RMSE and MAPE data of best sensor combination under two methods.

	Grey correlation degree	Consider average, standard, deviation, outlier and z-index
Optimal combination of three sensors	21,47,59	4,21,59
MAPE	1.79%	1.97%
RMSE	0.519	0.585

easily misses the optimal individual. It is appropriate to set the crossover rate to 0.8. If the probability of mutation is too small, the diversity of population will drop too fast, which will easily lead to unstable convergence. If the mutation rate is too large, although the diversity of population can be guaranteed, the probability of optimal solution will also increase. After debugging, the mutation rate is set to 0.06.

According to the flow of genetic algorithm with dual-structure coding, the program is programmed with MATLAB software. Taking the temperature change data of 13 sensors selected by correlation coefficient as input, and the gray correlation degree of the average change of sensor combinations and the average temperature change in the house under the same number as constraint conditions, 13 second best sensor combinations from 1 to 13 were obtained. Finally, the grey correlation degree between these 13 combinations and the temperature change trend in the laying hen house is calculated, and an optimal combination is obtained. Because the larger the fitness value of the sensor combination, the closer it is to the average temperature of the laying hen house. As shown in Fig. 7, it can be seen that when the number of sensors is 3, the grey correlation degree is the largest, and the corresponding optimal combination is the combination of 21, 47 and 59 sensors (Table 4), and their positions are shown in Fig. 8.

In order to verify the optimal sensor combination, mean absolute percentage error (MAPE) and root mean square error (RMSE) are used as statistical indicators. Another method is to map the average value, deviation, outlier and z- index of the combined error of each sensor with 13 numbers to the interval (0,1) and then add them. Then queue each sensor combination. The optimal sensor combination under the condition that the number is 3, the optimal sensor combination is (4,21,59) through iterative calculation of genetic algorithm of dual-structure coding. By calculating MAPE and RMSE of the two methods, it can be found that method of combining grey correlational degree with genetic algorithm proposed in this paper is better than some statistical index methods considering errors (Table 5). It shows that it is better to use grey correlational degree as a measurement index.

$$MAPE = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{h_i - y_i}{y_i}\right| \tag{6}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(h_i - y_i)^2} \tag{7}$$

here, n is the total number of data, h_i is the average temperature of the optimal combination at a specific time, and y_i is the average temperature in the laying hen house at a specific time.

V. CONCLUSION

In this paper, a layout optimization method of temperature sensors based on raw data is proposed. By simulating the actual hen house, a multi-layer cage hen house model is established, and the number and location of temperature sensors are optimized by applying the algorithm of combining genetic algorithm and grey correlational degree. Through example verification, the following conclusions are drawn:

(1) In the process of building the model of multi-layer cage laying hen house, considering the change of air temperature in the house with time and space, the optimized space is extended to three-dimensional space.

(2) In the actual multilayer cage laying hen house, the number of sensors is redundant. By calculating the correlation coefficient of temperature data collected by each sensor, the number of redundant sensors can be eliminated and the subsequent workload can be reduced. In this paper, for example, the number of temperature sensors decreased from 81 to 13, a decrease of 83.95%.

(3) Based on the results of eliminating redundant sensors, the sensor combination closest to the average temperature in the house is screened, and the sensor combination closest to the average temperature in the house is determined by applying the genetic algorithm of dual-structure coding and the grey correlational degree method. In this paper, for example, the number of temperature sensors has decreased from 13 to 3, with a decrease of 76.92%. Compared with MAPE and RMSE of another method, the effectiveness of grey relational degree method is proved.

(4) The layout optimization method of temperature sensors based on raw data greatly reduces the number of temperature sensors and the cost of modern farming.

Unfortunately, this study puts forward a method that can find the best sensor combination among a large number of combinations, and further research is needed for the optimization methods of the number and position of other types of sensors in the henhouse, such as humidity sensors in the henhouse and harmful gas concentration sensors, etc.

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